

Multi-Seed Benchmarking of Recurrent and Transformer Models for Indonesian Banking Stock Forecasting

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Abstract—This paper studies daily closing-price forecasting for major Indonesian banking stocks using recurrent sequence models and macroeconomic regressors under a repeated multi-seed protocol. The primary ablation compares LSTM and BiLSTM under two feature settings (`Without_Macro` and `With_Macro`) and two optimization modes (default vs. budgeted tuning via Genetic Algorithm), resulting in eight experimental cases per bank. Across seven banks, we apply Wilcoxon signed-rank, Diebold-Mariano, and SHAP-based analyses to reduce single-run bias and inspect model behavior. In the recurrent benchmark, BiLSTM configurations usually yield lower error than LSTM, with best-case RMSE improvements ranging from 4.91% to 35.66% for six out of seven banks. As an extension, we further evaluate Informer/Autoformer Transformer models and recurrent variants based on normalization, convolution, and temporal attention. Among the completed model families, the Transformer extension attains the lowest aggregate errors, with `informer_data` achieving RMSE 401.02, MAE 325.11, and MAPE 5.94%, while the recurrent variants do not exceed the original recurrent benchmark on average. These results are limited to the completed configurations and motivate further statistically matched validation.

Keywords—stock forecasting, LSTM, BiLSTM, Informer, Autoformer, macroeconomic features, genetic algorithm, Wilcoxon test, DM test, SHAP

I. INTRODUCTION

Forecasting bank stock prices remains challenging due to non-stationarity, regime shifts, and sensitivity to macroeconomic conditions. In emerging markets such as Indonesia, banking-sector prices are influenced not only by internal price dynamics but also by external drivers including interest rates, inflation, exchange rates, and broad market movements.

Recent work in financial time series forecasting has adopted deep learning models, especially recurrent networks and Transformer-based architectures, to capture temporal dependencies. However, many studies still provide limited ablation depth or weak statistical validation, which makes it difficult to judge whether reported gains persist across assets and random seeds.

This study builds a reproducible pipeline for Indonesian banking stocks that combines sequence modeling, macroeconomic regressors, repeated-seed evaluation, paired significance testing, and feature-attribution analysis. A central question is whether macroeconomic variables (`With_Macro`) improve on the price-only regime (`Without_Macro`) under the recurrent baseline, as of-

ten suggested in exogenous-variable forecasting literature [1], [2]. We also extend the comparison with Informer, Autoformer, and recurrent variants that add normalization, convolution, or temporal attention, and we use SHAP-based attributions where outputs are available to inspect feature relevance patterns [3].

The main contributions are as follows:

- A structured recurrent ablation design over model type (LSTM vs BiLSTM), feature regime (price-only vs price+macro), and optimization mode (off vs on).
- A multi-bank, multi-seed protocol for repeated performance estimation and variance analysis.
- Statistical validation using Wilcoxon signed-rank and Diebold-Mariano tests to assess significance of recurrent forecasting improvements.
- A SHAP-based feature-attribution analysis to quantify global feature relevance across banks, feature regimes, and optimization settings.
- A supplementary comparison against Informer/Autoformer and normalization, convolutional, and attention-based recurrent variants.

The remainder of this paper is organized as follows. Section II reviews related literature. Section III describes the dataset and preprocessing. Section IV presents the methodology. Section V details the experimental setup. Section VI discusses the results and significance tests. Section VII concludes and outlines future directions.

II. RELATED WORKS

Financial time-series forecasting has long used recurrent neural networks to model nonlinear temporal dependence in stock and index data. LSTM-type models remain practical baselines because they can encode sequential price behavior without requiring very large architectures. Recent studies also extend this family with bidirectionality, convolutional preprocessing, attention, or evolutionary feature construction, including CNN-BiLSTM-Attention models for stock-index prediction [4], Indonesian blue-chip stock comparisons with LSTM baselines [5], and LSTM variants combined with symbolic genetic programming for large-scale stock-return forecasting [6]. These works motivate using LSTM/BiLSTM as the primary recurrent benchmark while testing whether recurrent variants add value.

Transformer-based forecasting has become a second major direction for long and multivariate time series. Gen-

eral forecasting architectures such as Temporal Fusion Transformer (TFT) [7], Informer [8], Autoformer [9], and TimeXer [1] introduce attention, sequence distilling, decomposition, or exogenous-variable handling for temporal prediction. Finance-specific work has also adapted self-attention to stock selection and market forecasting, including Stockformer for price-volume factor modeling [10] and time-series Transformer models with SHAP/LIME analysis on BIST100 banking stocks [11]. These studies provide the basis for treating Informer and Autoformer as supplementary Transformer baselines rather than as replacements for recurrent statistical validation.

External variables are frequently used to enrich financial forecasting when their timing and economic relationship are informative. Prior work has studied macroeconomic indicators for Indonesian stock-market prediction [2], macro and sentiment features with LSTM-Attention [12], external indicators for bank-stock forecasting [13], and exogenous-variable Transformer designs [1], [14]. This literature supports an explicit comparison between *Without_Macro* and *With_Macro* regimes, while leaving open whether macro variables help uniformly across banks.

Interpretability methods are useful for checking which inputs a forecasting model relies on across assets, seeds, and feature regimes. SHAP provides a model-agnostic additive attribution framework [3] and has been applied to stock-market factor analysis and neural forecasting workflows [15]–[17]. Other recent financial-forecasting studies use LRP, dashboards, or broader explainable-deep-learning workflows [18]–[21]. In this study, SHAP is used as a feature-attribution layer for comparing model reliance patterns, not as a causal explanation of market movement.

Beyond architecture and features, applied financial forecasting often reports limited robustness checks. Single-run comparisons can obscure seed sensitivity, bank-specific behavior, and differences between average error improvements and paired statistical evidence. This paper addresses that gap by combining controlled ablation, repeated-seed evaluation, and paired statistical tests for the original recurrent benchmark, then reporting Transformer and recurrent-variant extensions with explicit caveats about the remaining cross-family validation.

III. DATASET AND PREPROCESSING

A. Dataset Description

This study focuses on seven Indonesian banking stocks: BBKA, BBNI, BBRI, BBTN, BDMN, BMRI, and BNGA. Daily market data include price and valuation-related variables, while macroeconomic variables include inflation-related indicators, policy rate, exchange rate, and market index features.

The target variable is the daily closing price. The analysis period follows a chronological split to avoid data leakage.

B. Feature Selection

The baseline feature set (*Without_Macro*) includes market micro and technical variables available in the bank-level dataset. The macro-augmented regime (*With_Macro*) extends this set with external variables:

- inflation rate and CPI-derived indicators,
- BI policy rate,
- USD/IDR exchange rate,
- IHSG index level and return proxy.

The final feature schema is fixed per regime before model training.

C. Feature Schema for SHAP Analysis

To support consistent SHAP analysis across model families, feature names are kept identical across banks and seeds, and the same temporal windowing process is used in every run. The *Without_Macro* regime uses 10 market/valuation features, while *With_Macro* adds 6 external variables (total 16 features). This fixed schema allows direct aggregation of feature attribution statistics for original recurrent and Transformer outputs, and it keeps the recurrent-variant outputs compatible with future SHAP analysis.

D. Data Preprocessing

We apply chronological sorting, forward-fill for missing observations, and standard scaling for both features and target during model training. Sequence windows are then constructed with a fixed lookback length.

Time split protocol:

- Training: 2014-01-02 to 2019-12-31
- Validation: 2020-01-02 to 2021-12-31
- Testing: 2022-01-02 to 2024-12-30

This split reflects strict temporal validation and supports fair out-of-sample evaluation.

IV. METHODOLOGY

A. Experimental Pipeline

The pipeline consists of five stages: data loading and scaling, sequence generation, model training, out-of-sample forecasting, and interpretability analysis. We evaluate multiple design factors using controlled ablation to isolate contribution from architecture, macro features, and optimization. Figure 1 summarizes the end-to-end workflow used in this study.

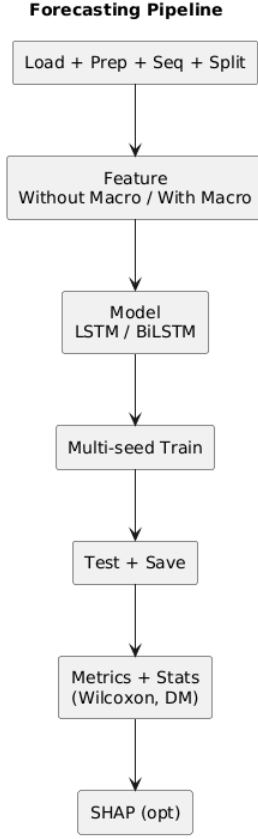


Fig. 1. Compact vertical pipeline overview used for the ablation and evaluation workflow.

B. Primary Recurrent Forecasting Framework

1) *LSTM and BiLSTM Baselines*: The original recurrent baselines use a single-layer recurrent encoder followed by a fully connected prediction head mapping the final hidden state to a scalar forecast. The shared architectural design ensures that performance differences are attributable to the directionality of temporal encoding rather than depth or capacity changes. The two variants are:

- **LSTM**: unidirectional temporal encoder with hidden size h , producing a single forward hidden state of dimension h .
- **BiLSTM**: bidirectional temporal encoder with hidden size h , concatenating forward and backward states to produce an output of dimension $2h$.

Both variants apply a dropout rate of 0.2 between the recurrent layer and the prediction head to mitigate overfitting. The prediction head consists of a single linear layer with ReLU activation, outputting the next-day closing price.

C. Supplementary Architecture Extensions

1) *Transformer Extension Models*: We extend the primary recurrent benchmark with Transformer-based sequence models adapted to one-step-ahead stock-price forecasting:

- **Informer**: an Informer-oriented encoder with positional encoding, ProbSparse-style attention, and sequence distilling to reduce long-sequence computation [8].

- **Autoformer**: an Autoformer-oriented encoder using moving-average decomposition and an Auto-Correlation-style temporal block to separate trend and seasonal components [9].

For reproducibility, the implementation also keeps lighter Informer-style and Autoformer-style baselines in the code-base. These Lite variants are not the primary Transformer results because they omit the original ProbSparse attention and Auto-Correlation mechanisms.

2) *Recurrent Architecture Variants*: To test whether the recurrent family can be strengthened without switching to Transformer models, we add ten recurrent variants: LSTM/BiLSTM with pre-normalization, post-normalization, and combined pre+post normalization; CNN-LSTM and CNN-BiLSTM encoders; and LSTM/BiLSTM with temporal attention. The normalization variants apply LayerNorm before and/or after recurrent encoding. The CNN variants apply a one-dimensional convolutional feature extractor before recurrent modeling. The attention variants learn temporal weights over hidden states before the prediction head.

D. Budgeted Hyperparameter Optimization via Genetic Algorithm

To isolate the effect of hyperparameter tuning from architectural choices, we introduce a constrained search procedure based on a Genetic Algorithm (GA). The GA operates on a fixed computational budget and optimizes three hyperparameters using validation loss as the fitness objective:

- **Hidden dimension (h)**: searched over $\{32, 64, 128, 256\}$,
- **Learning rate (α)**: searched over $\{1 \times 10^{-4}, 5 \times 10^{-4}, 1 \times 10^{-3}, 5 \times 10^{-3}\}$,
- **Sequence length (L)**: searched over $\{10, 20, 30, 60\}$.

Each GA individual encodes one (h, α, L) triplet. The population size is set to 8 and the number of generations is fixed at 5, yielding a total evaluation budget of $8 \times 5 = 40$ configurations per case. Tournament selection (size 3) is used for parent selection, and single-point crossover with a per-gene mutation probability of 0.2 is applied. The hand-picked default configuration used in non-GA cases is $(h = 64, \alpha = 10^{-3}, L = 20)$. Configurations tagged as “_ga” in the results tables indicate the GA-optimized setting. This design tests whether budgeted tuning improves accuracy and stability versus fixed defaults, acting as an ablation factor for optimization effort rather than a contribution claim about the search algorithm itself.

For the supplementary Transformer experiments, the code also supports a budgeted random-search mode. This mode uses the same validation objective and a small candidate budget to generate the `_random` Transformer cases and fast baseline estimates; it is separate from the GA workflow used for recurrent-family tuning and should not be read as a completed Transformer GA sweep.

E. Statistical Validation

Beyond average error reporting, we apply two complementary tests:

- **Wilcoxon signed-rank test:** a nonparametric paired test applied on seed-level RMSE vectors ($n = 5$) to assess whether median performance differences are significant without assuming normality.
- **Diebold-Mariano (DM) test:** applied on aligned forecast-error residuals to evaluate whether one model produces significantly lower prediction errors than another.

The Wilcoxon test provides seed-level robustness evidence, while the DM test assesses time-series-level significance. Agreement between both tests strengthens confidence in reported gains.

V. EXPERIMENTAL SETUP

A. Evaluation Metrics

Primary metrics are MSE, RMSE, MAE, and MAPE on the test partition. We report both point estimates (mean) and seed-wise variability (standard deviation) across five independent runs to characterize stability.

B. Primary Recurrent Benchmark

The main ablation factors are:

- **Model family:** LSTM and BiLSTM
- **Feature regime:** Without_Macro (10 features) vs With_Macro (16 features)
- **Optimization mode:** default ($h=64, \alpha=10^{-3}, L=20$) vs GA-optimized (“_ga”)

This yields $2 \times 2 \times 2 = 8$ recurrent-model cases per bank for controlled comparison, or 280 recurrent runs across seven banks and five seeds.

C. Supplementary Extension Experiments

Additional experiments evaluate two architecture-extension groups using the same bank and seed protocol. The Transformer extension contains Informer and Autoformer cases under price-only and macro regimes, including fast random-search variants, for 280 runs. The recurrent-variant extension contains normalization, convolutional recurrent, and attention-augmented LSTM/BiLSTM variants, for 1400 runs. These runs are reported as supplementary evidence because paired statistical tests across all model families have not yet been regenerated.

D. Multi-Seed Protocol

Each configuration is executed with five seeds: $\{42, 52, 62, 72, 82\}$. The complete run table stores configuration metadata, per-epoch metrics, and file paths for downstream significance analysis. All cases use identical data splits and seed sets to ensure fairness.

E. Implementation Details

Implementation is based on Python and PyTorch. Training uses the Adam optimizer with ReduceLROnPlateau scheduling and validation-based early stopping. The recurrent baseline uses batch size 32 in the original study; GPU-accelerated Transformer and recurrent-variant sweeps use larger batch configurations where supported by memory. The forecasting horizon is one day. Code and outputs are

TABLE I
GLOBAL BEST CASE BY MODEL FAMILY (MEAN OVER 7 BANKS AND 5 SEEDS).

Family	Best case	RMSE	MAE	MAPE (%)
Original recurrent	bilstm_data_ga	680.83	591.04	10.65
Recurrent variants	cnn_bilstm_data_ga	754.96	668.21	12.35
Transformer	informer_data	401.02	325.11	5.94

TABLE II
BEST CASE PER BANK ACROSS ORIGINAL RECURRENT, RECURRENT VARIANTS, AND TRANSFORMERS.

Bank	Original recurrent	Recurrent variant	Transformer
BBCA	bilstm_data (1881.53)	bilstm_attention_data (2004.10)	autoformer_macro (545.93)
BBNI	bilstm_data (335.85)	cnn_bilstm_data (362.88)	informer_data (219.66)
BBRI	bilstm_data (525.70)	bilstm_attention_data (580.65)	informer_macro (379.21)
BBTN	bilstm_data_ga (51.44)	bilstm_attention_data (70.84)	informer_data_random (43.01)
BDMN	lstm_data (120.72)	cnn_bilstm_data_ga (116.73)	informer_data (76.82)
BMRI	bilstm_macro_ga (1388.71)	cnn_bilstm_macro (1533.72)	informer_data_random (885.21)
BNGA	bilstm_data (244.14)	cnn_bilstm_data_ga (247.60)	informer_data (132.56)

organized under the project repository, with raw experiment artifacts stored in `code/results_study_full`, `code/results_transformer_study`, and `code/results_recurrent_variants_study`.

F. Cross-Model Fairness Protocol

To maintain fairness, all reported families use identical bank sets, chronological splits, and seed sets. Feature regimes are matched as price-only or macro-augmented. Optimization budgets are kept budgeted rather than exhaustive, so results should be interpreted as controlled experimental evidence rather than a claim of globally optimal hyperparameters.

VI. RESULTS AND DISCUSSION

A. Performance Comparison Across Model Families

Table I summarizes the strongest global case from each completed experiment family. The Transformer family shows the lowest observed aggregate error among the completed runs, with `informer_data` reducing mean RMSE from 680.83 in the best original recurrent case to 401.02. The recurrent-variant sweep expands the architecture search space, but its best global case does not surpass the original recurrent winner.

The same pattern appears at bank level among the completed runs. Table II compares the best original recurrent, recurrent-variant, and Transformer case for each bank. Transformer models win all seven bank-level comparisons in this completed set. The average best-per-bank improvements over the original recurrent winners are 38.31% for RMSE, 42.17% for MAE, and 41.95% for MAPE.

B. Statistical Significance Analysis

Table III reports the inferential validation currently available for the original recurrent benchmark, comparing each bank’s best recurrent case against `lstm_data`. RMSE gains are positive in six banks, ranging from 4.91% to 35.66%. For the Transformer extension, these values should be read as the lowest observed errors among completed runs; cross-family Wilcoxon and Diebold-Mariano tests should be regenerated before making any formal significance claim.

Table IV presents a cross-family ranking of the strongest global cases. The top eight positions are all Transformer

TABLE III
ORIGINAL RECURRENT SIGNIFICANCE SUMMARY AGAINST
LSTM_DATA.

Bank	Best case	Improve (%)	Wilcoxon p	DM p
BBCA	bilstm_data	8.76	0.4375	2.28×10^{-13}
BBNI	bilstm_data	10.47	0.0625	4.84×10^{-11}
BBRI	bilstm_data	28.23	0.3125	$< 10^{-16}$
BBTN	bilstm_data_ga	35.66	0.3125	2.22×10^{-16}
BDMN	lstm_data	0.00	1.0000	n/a
BMRI	bilstm_macro_ga	11.18	0.4375	2.23×10^{-11}
BNGA	bilstm_data	4.91	0.3125	4.78×10^{-5}

TABLE IV
CROSS-FAMILY GLOBAL RANKING ACROSS ALL BANKS (LOWER IS
BETTER).

Family	Case	RMSE	MAE	MAPE (%)
Transformer	informer_data	401.02	325.11	5.94
Transformer	informer_data_random	421.00	342.67	6.67
Transformer	informer_macro	450.05	385.37	7.59
Transformer	autoformer_data	453.87	388.04	9.49
Transformer	autoformer_macro	496.49	420.89	10.61
Transformer	autoformer_data_random	501.79	435.21	10.31
Original recurrent	bilstm_data_ga	680.83	591.04	10.65
Original recurrent	bilstm_data	681.26	589.76	10.74
Recurrent variants	cnn_bilstm_data_ga	754.96	668.21	12.35

cases; the best original recurrent case appears next, and the strongest recurrent variant follows after the original recurrent winners. This is evidence that the new Transformer models achieve the lowest observed aggregate errors among the completed runs, but it is not a formal cross-family significance result.

To directly answer whether macroeconomic feature engineering improves accuracy, Table V summarizes how often macro-augmented cases win within each family. Macro features are not uniformly beneficial. The global winner in every family is data-only, although macro cases remain useful in specific banks such as BMRI for original recurrent and recurrent variants, and BBBCA/BBRI for Transformers.

This evidence suggests that macro variables are useful only in specific regimes/banks rather than universally. This pattern is compatible with the broader literature: methods designed for exogenous variables can improve forecasting when the external signals are informative and properly aligned [1], [14], and multiple IEEE studies also report gains from macro-integrated forecasting pipelines [2], [12], [13].

C. XAI Analysis with SHAP

We generated SHAP feature-attribution outputs for all 280 original recurrent checkpoints (7 banks $\times$ 8 cases $\times$ 5 seeds). This produced 3,640 feature-attribution rows over 16 unique features. Using global top-1 frequency, the most frequent top-ranked features are *Low Price* (39 runs), *Bid Price* (31), *Price Change 1 Day Percent* (28), and *Price to Sales Ratio* (28). The pattern indicates that short-horizon market microstructure and valuation factors appear more often than macro indicators in the feature attributions.

Because a small number of runs showed extreme SHAP magnitudes (maximum $\approx 1.51 \times 10^{13}$), we also report a robust global ranking using a 99th-percentile cap. Under this robust aggregation, the top features remain stable: *Low Price* (0.00465), *Bid Price* (0.00428), *Price Change 1 Day Percent*

TABLE V
MACRO-FEATURE REGIME SUMMARY ACROSS MODEL FAMILIES.

Family	Global best regime	Macro bank wins	Best macro bank(s)
Original recurrent	Data-only	1/7	BMRI
Recurrent variants	Data-only	1/7	BMRI
Transformer	Data-only	2/7	BBCA, BBRI

TABLE VI
GLOBAL SHAP SUMMARY (ROBUST AGGREGATE OVER ALL 280
RECURRENT RUNS).

Feature	Mean —SHAP— (cap99)	Top-1 count
Low Price	0.00465	39
Bid Price	0.00428	31
Price Change 1 Day Percent	0.00407	28
High Price	0.00394	25
Moving Average 20 Day	0.00383	25
Price to Book Ratio	0.00383	25
Price Earnings Ratio	0.00365	20
Price to Sales Ratio	0.00362	28

(0.00407), *High Price* (0.00394), and *Moving Average 20 Day* (0.00383). Hence, the main interpretation does not depend on a few outlier attributions.

D. Supplementary Transformer Extension

As shown in Tables I and II, the Transformer sweep records the lowest observed errors among the completed extension runs. *Informer_data* is the best global case, while the bank-level winners are distributed across *informer_data*, *informer_data_random*, *informer_macro*, and *autoformer_macro*. The *_random* entries come from the budgeted Transformer random-search mode described in the methodology, not from GA-optimized Transformer runs. This distribution indicates that both architecture choice and data regime remain bank-dependent even though the Transformer family has the lowest observed aggregate error. These results support further validation of Informer and Autoformer modeling, but they are reported as extension evidence because paired statistical tests against the recurrent winners have not yet been regenerated.

E. Supplementary Recurrent-Variant Extension

The recurrent-variant sweep contains 1400 runs over normalization, convolutional recurrent, and attention-augmented LSTM/BiLSTM cases. The strongest global variant is *cnn_bilstm_data_ga*, with RMSE 754.96, MAE 668.21, and MAPE 12.35%. The top three recurrent variants are *cnn_bilstm_data_ga*, *cnn_bilstm_data*, and *cnn_lstm_data*, indicating that convolutional preprocessing is the most competitive recurrent-variant subgroup.

However, recurrent variants do not improve the original recurrent benchmark on average. Their average best-per-bank differences relative to the original recurrent winners are -10.18% RMSE, -7.90% MAE, and -7.41% MAPE, where negative values indicate worse performance. The only bank-level exception is BDMN, where *cnn_bilstm_data_ga* slightly improves RMSE over *lstm_data* (116.73 vs. 120.72). Recurrent variants are therefore useful as architecture-ablation evidence, but they do not replace the original BiLSTM baseline in this study.

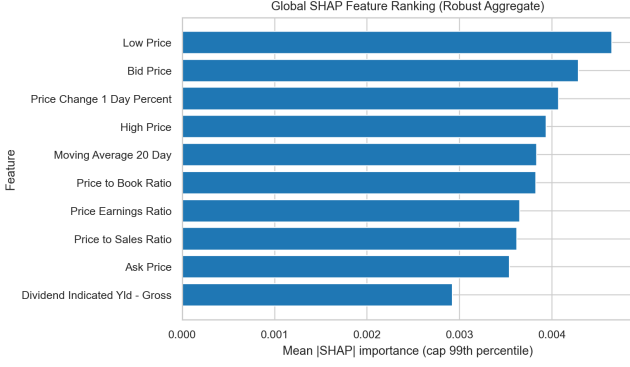


Fig. 2. Global SHAP ranking using robust mean importance (99th-percentile capped).

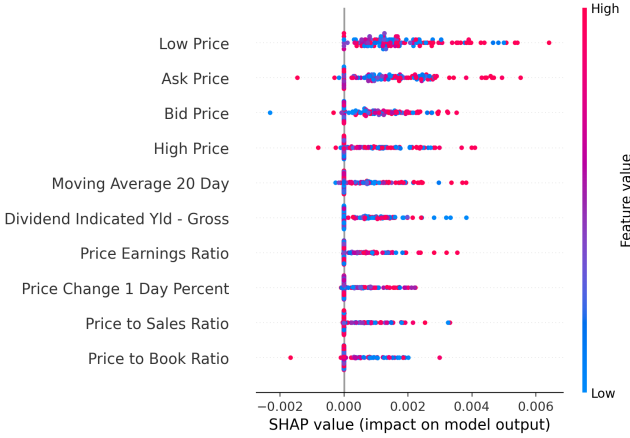


Fig. 3. SHAP beeswarm plot for the best-performing recurrent checkpoint (BBTN, BiLSTM_Data_GA, seed 62).

F. Discussion and Threats to Validity

Three validity points are important. First, the chronological split is fixed; different split boundaries may change relative ranking. Second, seed-level Wilcoxon tests are underpowered with $n = 5$, which explains why several improvements with very small DM p -values are not mirrored by Wilcoxon significance. Third, optimization benefits are heterogeneous across banks, indicating risk of overfitting when tuning is not controlled.

The extension experiments add two additional caveats. Transformer and original recurrent runs have SHAP outputs, but recurrent variants were executed without SHAP to control compute cost. In addition, the Transformer and recurrent-variant results currently provide multi-seed aggregate evidence, while the strongest inferential validation remains the original recurrent Wilcoxon and DM analysis. Future work should regenerate paired statistical tests across the best recurrent, Transformer, and recurrent-variant cases before making cross-family significance claims.

VII. CONCLUSION

This paper presents a reproducible experimental pipeline for forecasting Indonesian bank stock prices using recurrent sequence models, macroeconomic features, multi-seed evaluation, feature-attribution outputs, and paired statistical testing.

Within the original recurrent benchmark, BiLSTM-based settings generally outperform LSTM across the evaluated banks, with best-case RMSE gains over the `lstm_data` baseline reaching up to 35.66%. Diebold-Mariano tests indicate significant forecast-error reduction in six banks, while Wilcoxon results are less conclusive due to the five-seed sample. Macro features and budgeted tuning are not uniformly beneficial, so the evidence favors bank-specific model selection rather than one fixed configuration for all emittents.

The supplementary architecture experiments change the follow-up direction rather than the validation status of the original benchmark. Informer/Autoformer runs produce the lowest observed aggregate errors among the completed families, with `informer_data` reaching RMSE 401.02, MAE 325.11, and MAPE 5.94%. Recurrent variants based on normalization, convolution, and attention broaden the ablation space, but they do not improve the original recurrent baseline on average. Thus, Transformers are the strongest candidate for paired cross-family validation, while recurrent variants are better interpreted as architecture-ablation evidence.

Several limitations should be noted. First, the evaluation is restricted to seven Indonesian banking stocks, which constrains generalizability to other markets and sectors. Second, the five-seed protocol limits statistical power for nonparametric tests. Third, Transformer and recurrent-variant statistical tests should be regenerated from raw prediction outputs to match the original recurrent validation depth. Fourth, recurrent variants currently lack SHAP outputs because the 1400-run sweep was executed without SHAP for compute-cost control.

Immediate future work includes: (i) extending validation to walk-forward protocols, (ii) increasing seed count for stronger nonparametric testing power, (iii) ablation on feature lag to investigate whether temporal misalignment explains macro feature failures, and (iv) paired statistical validation and targeted SHAP analysis across the best recurrent, Transformer, and recurrent-variant cases.

REFERENCES

- [1] Y. Wang, H. Wu, J. Dong, G. Qin, H. Zhang, Y. Liu, Y. Qiu, J. Wang, and M. Long, "Timexer: Empowering transformers for time series forecasting with exogenous variables," *arXiv preprint arXiv:2402.19072*, 2024. [Online]. Available: <https://arxiv.org/abs/2402.19072>
- [2] S. F. Adyatma and A. Alamsyah, "The indonesia stock exchange composite prediction based on macroeconomic indicators using arima, lstm, and ann," in *2022 8th International Conference on Science and Technology (ICST)*, 2022.
- [3] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2017/hash/8a20a8621978632d76c43dfd28b67767-Abstract.html
- [4] Y. Yang and J. Xu, "Stock index prediction based on cnn-bilstm-attention," in *2024 IEEE 2nd International Conference on Electrical, Automation and Computer Engineering (ICEACE)*, 2024.
- [5] Y. Saleh, M. Pradana, and B. R. Kartawinata, "Comparative forecasting of indonesian blue-chip stocks using arima, ann and lstm models," in *2025 IEEE 16th Control and System Graduate Research Colloquium (ICSGRC)*, 2025.

- [6] Q. Li, N. Kamaruddin, S. S. Yuhaniz, and H. A. A. Al-Jaifi, "Forecasting stock prices changes using long-short term memory neural network with symbolic genetic programming," *Scientific Reports*, vol. 14, no. 422, 2024.
- [7] B. Lim, S. O. Arik, N. Loeff, and T. Pfister, "Temporal fusion transformers for interpretable multi-horizon time series forecasting," *arXiv preprint arXiv:1912.09363*, 2019. [Online]. Available: <https://arxiv.org/abs/1912.09363>
- [8] H. Zhou, S. Zhang, J. Peng, S. Zhang, J. Li, H. Xiong, and W. Zhang, "Informer: Beyond efficient transformer for long sequence time-series forecasting," *arXiv preprint arXiv:2012.07436*, 2020. [Online]. Available: <https://arxiv.org/abs/2012.07436>
- [9] H. Wu, J. Xu, J. Wang, and M. Long, "Autoformer: Decomposition transformers with auto-correlation for long-term series forecasting," *arXiv preprint arXiv:2106.13008*, 2021. [Online]. Available: <https://arxiv.org/abs/2106.13008>
- [10] B. Ma, Y. Xue, Y. Lu, and J. Chen, "Stockformer: A price-volume factor stock selection model based on wavelet transform and multi-task self-attention networks," *arXiv preprint arXiv:2401.06139*, 2024. [Online]. Available: <https://arxiv.org/abs/2401.06139>
- [11] S. S. Calik, A. Akyuz, Z. H. Kilimci, and K. Colak, "Explainable-ai powered stock price prediction using time series transformers: A case study on bist100," *arXiv preprint arXiv:2506.06345*, 2025. [Online]. Available: <https://arxiv.org/abs/2506.06345>
- [12] S. S. Begum, N. S. Fatima, and M. Abbas, "Enhancing stock market prediction with an lstm-attention model integrating macroeconomic and sentiment features," in *2024 International Conference on Innovative Computing, Intelligent Communication and Smart Electrical Systems (ICES)*, 2024.
- [13] J. Dechjijaruwate and C. Pisarn, "Forecasting stock prices of commercial banks in the stock exchange of thailand using machine learning," in *2025 20th International Joint Symposium on Artificial Intelligence and Natural Language Processing (iSAI-NLP)*, 2025.
- [14] X. Qiu, Y. Zhu, Z. Li, X. Wu, B. Yang, and J. Hu, "Dag: A dual correlation network for time series forecasting with exogenous variables," *arXiv preprint arXiv:2509.14933*, 2025. [Online]. Available: <https://arxiv.org/abs/2509.14933>
- [15] C. M. Shih and K. C. Yang, "Using lightgbm + shap to analyze the impact of factors on stock market investment," in *2024 IEEE International Conference on Industrial Engineering and Engineering Management (IEEM)*, 2024.
- [16] M. H. Begum, F. M. A. Rahman, M. A. K. Azad, C. R. Debi, K. M. M. Uddin, and A. K. M. Masum, "Diverse minds think alike: A shap-guided rnn for stock price forecasting with smart sma optimization," in *2025 3rd International Conference on Artificial Intelligence, Blockchain, and Internet of Things (AIBThings)*, 2025.
- [17] Y. Gupta S, P. Naresh, C. Nagesh, P. Praveen, N. Kirankumar, and K. Surendra, "An interpretable framework for stock market forecasting using long short-term memory (lstm) networks with shap - explainable ai (xai) method," in *2025 6th International Conference on Data Intelligence and Cognitive Informatics (ICDICI)*, 2025.
- [18] H. Tanveer, S. Arshad, H. Ameer, and S. Latif, "Interpretability in financial forecasting: The role of explainable ai in stock market," in *2024 14th International Conference on Software Technology and Engineering (ICSTE)*, 2024.
- [19] H. Tanveer and S. Latif, "Decoding stock market predictions: Insights from explainable ai using layer-wise relevance propagation," in *2024 26th International Multi-Topic Conference (INMIC)*, 2024.
- [20] J. Jagannathan and V. V. S. K. Goditi, "Integration of a hybrid model with xai for stock price forecasting: A dashboard-driven approach," in *2024 International Conference on IT Innovation and Knowledge Discovery (ITIKD)*, 2025.
- [21] S. Ayyappan, A. A. Gadgil, R. Gupta, H. B. Furaiji, S. Najah, and J. A. Jasim, "Time series forecasting framework using explainable deep learning for stock price prediction," in *2025 International Conference on Recent Innovation in Science Engineering and Technology (ICRISET)*, 2025.